Proof by Contradiction

Questions 1–18

Question 1. For each of the following statements, write the negation:

a) The cat is black.
b) Some cats are black.
c) Some cats are not black.
d) Line A is parallel to line B.
e) The probability has increased.
f) $x \geq 0$
g) $x = 2$
h) a is irrational.
i) n is even and prime.

Question 2. For each of the following statements, write the negation:

a) For all integers, n, either n is positive or n is negative.
b) n is a number with at least 3 distinct prime factors.
c) Every prime number is odd.
d) The sum of a rational number and an irrational number is always irrational.
e) If n^2 is even, then n is even.

Question 3. A student is attempting to prove by contradiction that if p and q are positive integers, that $\frac{p}{q} + \frac{q}{p} \geq 2$ Their first line of working states:

> Given that p and q are negative numbers, $\frac{p}{q} + \frac{q}{p} \geq 2$

Correct their first line.

Question 4. Prove that the following numbers are irrational:

a) $\sqrt{2}$
b) $\sqrt{3}$
c) $\sqrt{5}$

Question 5. Prove that $\sqrt[3]{5}$ is an irrational number.

Question 6. Prove that $1 + \sqrt{2}$ is irrational.

Question 7. Prove that if $x^2 - 5x + 6 < 0$ then $2 < x < 3$

Question 8. Prove that there is no greatest multiple of 3

Question 9. Prove that there are infinitely many prime numbers.

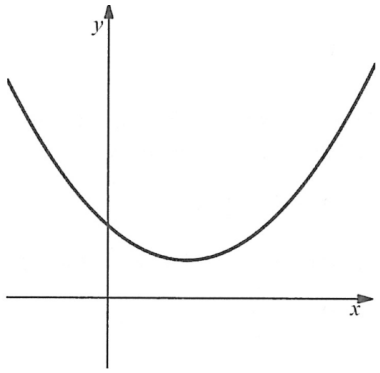
Question 10. Prove that the difference between a rational number and an irrational number is always irrational.

Question 11.

a) Prove that the product of a positive rational number and a positive irrational number is irrational

b) Hence, state the values for which it is not true that the product of a rational and an irrational number is irrational.

Question 12. Lynn has sketched the graph of $f(x) = kx^2 - 3kx + 8$ which can be seen below.



She claims that the values of k for which this is the correct sketch are $0 < k < \frac{32}{9}$
Use proof by contradiction to prove her claim.

Question 13. Prove that for all $x > 0$, $x + \frac{1}{x} \geq 2$

Question 14. Prove that for any real value of a and b then $a^2 + b^2 > 2ab$

Question 15. Prove that $\log_{10} 3$ is an irrational number.

Question 16.

a) Show that if $p^2 - 6q = 5$ then p^2 is odd

b) Hence, prove by contradiction that there are no integers a and b such that $a^2 - 6b = 5$

Question 17. Prove that 3 is the only prime number which is 1 less than a square number.

Question 18. Prerequisite knowledge for Q18: Double Angle Formula Prove that for all real θ, $\cos\theta+\sin\theta \leq \sqrt{2}$